

TASK 5

Write a C/JAVA program on Rivest Cipher 4(RC4) logic.

Aim:To implement RC4 LOGIC

Program:

```
import java.security.*;
import java.io.*;
import java.util.*;
import javax.crypto.*;

public class Task5 {

    static Scanner s = new Scanner(System.in);

    private static String encrypt(String data, String key)
    {

        int[] s = new int[256];

        int[] k = new
        int[256];

        int temp, i, j = 0, z;

        StringBuilder encryptedData = new StringBuilder();

        char[] dataChars =
        data.toCharArray(); char[]
        keyChars = key.toCharArray();
        int[] dataInt = new
        int[data.length()]; int[] keyInt =
        new int[key.length()];

        for (int m = 0; m < data.length(); m++)
        {

            dataInt[m] = (int) dataChars[m];

        }

        for (int m = 0; m < key.length(); m++)
        {

            keyInt[m] = (int) keyChars[m];

        }

        for (int m = 0; m < 256; m++)
        {
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        s[m] = m;
        k[m] = keyInt[m % key.length()];
    }

    j = 0;
    for (int m = 0; m < 256; m++)
    {
        j = (j + s[m] + k[m]) % 256;
        temp = s[m];
        s[m] = s[j];
        s[j] = temp;
    }

    i = 0;
    j = 0;
    for (int l = 0; l < data.length(); l++)
    {
        i = (i + 1) % 256;
        j = (j + s[i]) % 256;

        temp =
            s[i]; s[i] =
            s[j]; s[j] =
            temp;

        z = s[(s[i] + s[j]) % 256];
        encryptedData.append((char) (z ^ dataInt[l]));
    }

    return encryptedData.toString();
}

private static String decrypt(String data, String
    key) { return encrypt(data, key);
}

public static void main(String[] args) {
    System.out.println("Enter the plain
text:"); String data = s.nextLine();
    System.out.println("Enter the key:");

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String key = s.nextLine();

String encryptedData = encrypt(data, key);

System.out.println("Encrypted: " + encryptedData);

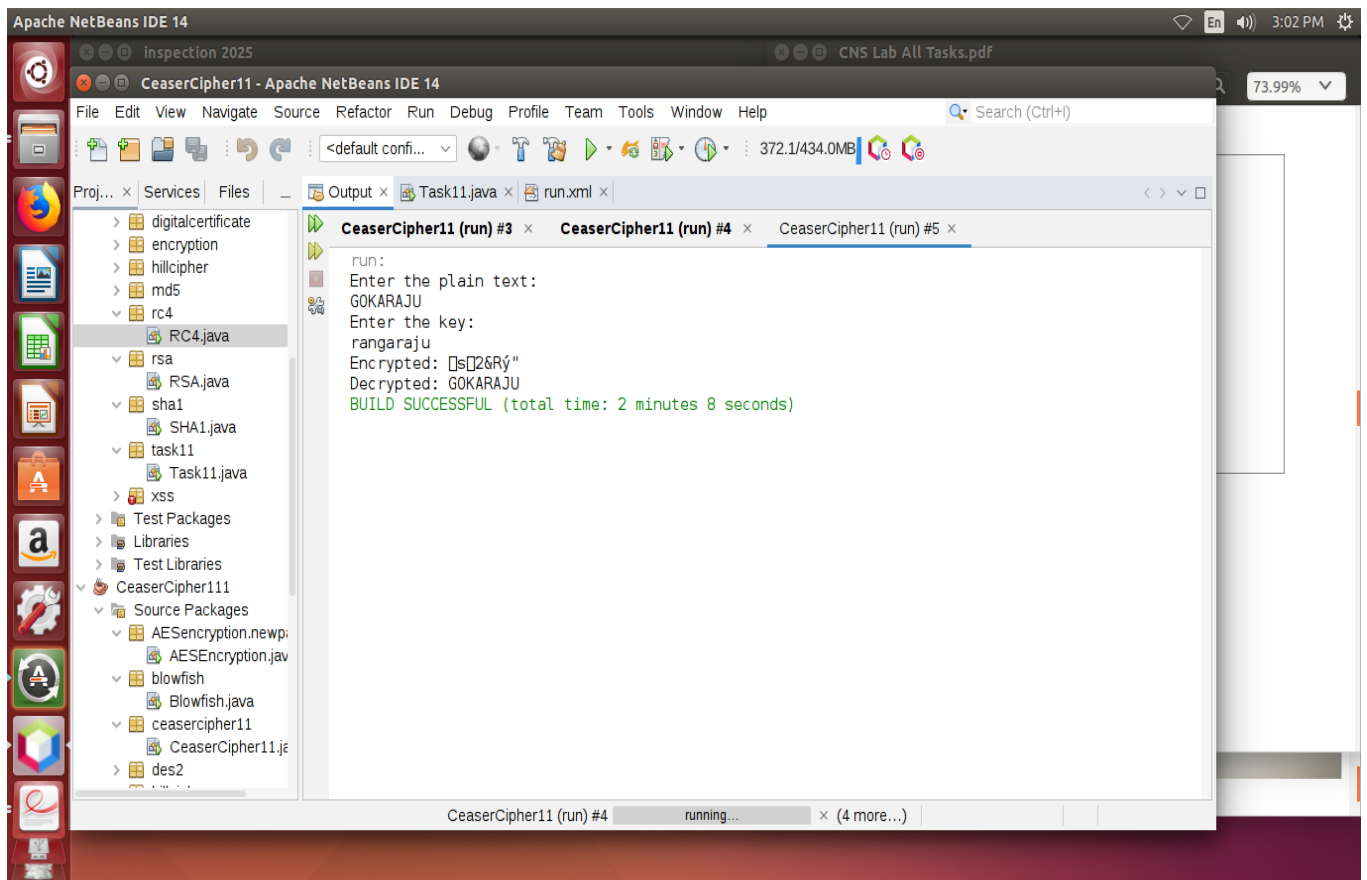
System.out.println("Decrypted: " + decrypt(encryptedData, key));

}

}

```

OUTPUT:



VIVA QUESTIONS

1. What is the RC4 algorithm, and how does it work?
2. What is the role of the key-scheduling algorithm (KSA) in RC4?
3. What are the major vulnerabilities of RC4, and why is it considered insecure today?
4. Where was RC4 commonly used, and what algorithms have replaced it in modern cryptography?